Segerstrom High School
2025 - 2026

AP Computer Science A
A Comprehensive Self-Study Guide Outline

Context

What is AP Computer Science A?

AP Computer Science A (APCSA), often regarded as the predecessor of AP Computer Science Principles (APCSP), offers a more in-depth computer science course on the **Java programming language**. It is a college-level introductory course that doesn't require any previous programming experience. That being said, **APCSA is generally considered more challenging than APCSP**. It heavily relies on logic-based problems and Object-Oriented Programming (OOP), two components that are foundational for Java. Additionally, APCSA covers data handling techniques such as searching and sorting algorithms, recursion, inheritance, and array algorithms. In essence, **APCSA delves more into theory, while APCSP is more focused on applications of computer science.**

Why should I take APCSA? Am I ready for it?

APCSA's main benefit is **college credit**. Not only are you self-studying an AP class (which already demonstrates above-and-beyond effort), APCSA is typically given credit for a CS-introductory course at a much higher rate than APCSP.

That being said, **APCSA isn't for everybody.** Although you technically don't need any previous experience, this guide is formatted in a way which **expects you to have either already taken APCSP, or are currently taking it.** Additionally, there wouldn't be any point of passing the APCSA exam unless you are planning on going into a STEM field. We highly suggest you talk to Mr. Schultz and Mr. Werner if you are planning on self-studying APCSA.

Resources

- CollegeBoard - AP Computer Science A
    - Official website for APCSA
    - APCSA 2025 Course Exam and Description
        - You don't have to read through everything, but please skim through the unit contents and the format of the exam. It's important to check the unit contents that way you can checklist all of the things you've learned as you progress throughout this course.
    - APCSA Past FRQs
        - Extremely important you practice with past FRQs and self-grade yourself. The more FRQs you do, the more accustomed you'll be into what the graders are strict (or not) on.
    - APCSA 2025 Revisions
        - I highly recommend reading the revisions, the most important information is that inheritance will no longer be tested, and a new unit (file reading) will be added.
    - APCSA Java Quick Reference
        - Provided cheat sheet in the exam.
- CSAwesome - 2025/2026
    - Main textbook of this course and what the timeline will be based on.
    - CSAwesome - APCSA Notebook
        - Notebook that has main ideas/summary notes.
- Barron's AP CSA Review - 13th Edition (2025-2026)
    - Preparation book used to review FRQs and general exams. I highly recommend buying this 1-2 months prior to the exam.
    - If you do not have the resources to buy this book, please email me at pablo.silvalizarraga@gmail.com

Timeline

Week 1 - Getting Started

Monday 8/11

- Get access to AP Classroom (contact Mr. Werner)
- 0.1 - Preface
  - No need to take notes.
- 0.2 - About the AP CSA Exam
  - Read carefully, take notes.

Tuesday 8/12

- 0.3 - Transitioning from AP CSP to AP CSA
  - Read carefully, take notes, and read Java code from the google documents.
- 0.4 - Java Development Environments
  - Set your environment up either at home or in Mr. Schultz's classroom.

Wednesday 8/13

- 0.5 - Growth Mindset and Pair Programming
  - Watch videos, take notes, and select your partner for AP CSA.
- 0.6 - Short PD Pretest
  - Take the test, note mistakes.
  - Not meant to score well.

Thursday 8/14

- 0.7 - Pretest for the AP CSA Exam
  - Take the test, note mistakes.
  - Not meant to score well.

Friday 8/15

- 0.8 - Survey
  - Answer truthfully.

Week 2 - Introduction to Programming

Monday 8/18 - Delayed Start

- 1.1 - Introduction to Algorithms, Programming, and Compilers
    - Take notes until Activity 1.1.6
    - Make sure to complete all activities as you go.

Tuesday 8/19

- 1.1 - Introduction to Algorithms, Programming, and Compilers
    - Take notes until you finish 1.1
    - Complete activities as you go

Wednesday 8/20

- 1.2 - Variables and Data Types
    - Take detailed notes, focus on activities.

Thursday 8/21

- 1.3 - Expressions and Output
    - Take detailed notes, focus on activities.

Friday 8/22

- 1.4 - Assignment and Input
    - Take detailed notes, focus on activities.
    - Review notes if necessary throughout the weekend.

Week 3 - More on Data Manipulation

Monday 8/25 - Delayed Start

- 1.5 - Casting and Ranges of Values
    - Take notes until Topic 1.5.2
    - Complete activities as you go

Tuesday 8/26

- 1.5 - Casting and Ranges of Values
    - Take notes until you finish 1.5

Wednesday 8/27

- 1.6 - Compound Assignment Operators
    - Take detailed notes, focus on activities.

Thursday 8/28

- Unit 1 Part A Review
    - Unit 1a Summary
    - Unit 1a Practice Toggle Coding

Friday 8/29

- Unit 1 Part A Review
    - Unit 1a Practice Coding
    - Unit 1a Multiple-Choice Exercises
        - Test yourself, avoid using outside resources.

Week 4 - Introduction to Classes

Monday 9/1 - No School

Tuesday 9/2

- 1.7 - APIs and Libraries
    - Take detailed notes, focus on activities.

Wednesday 9/3

- 1.8 - Documentation with Comments and Preconditions
    - Take detailed notes, focus on activities.

Thursday 9/4

- 1.9 - Method Signatures
    - Take detailed notes, focus on activities.

Friday 9/5

- 1.10 - Calling Class Methods
    - Take detailed notes, focus on activities.

Week 5 - Classes and Strings

Monday 9/8 - Delayed Start

- 1.11 - Using the Math Class
    - Take notes until Topic 1.11.2
    - Complete activities as you go

Tuesday 9/9

- 1.11 - Using the Math Class

    - Take detailed notes, focus on activities.

Wednesday 9/10

- 1.12 - Objects - Instances of Classes
    - Take detailed notes, focus on activities.

Thursday 9/11

- 1.13 - Creating and Initializing Objects: Constructors
    - Take detailed notes, focus on activities.

Friday 9/12

- 1.14 - Calling Instance Methods
    - Take detailed notes, focus on activities.
    - Review notes if necessary throughout the weekend.

Week 6 - Review for Unit 1

Monday 9/15

- 1.15 - Strings
    - Take detailed notes, focus on activities.

Tuesday 9/16

- Unit 1 Part B Review
    - Unit 1b Summary
    - Unit 1b Practice Toggle Coding

Wednesday 9/17

- Unit 1 Part B Review
    - Unit 1b Practice Coding
    - Unit 1b Multiple-Choice Exercises
        - Test yourself, avoid using outside resources.

Thursday 9/18

- 1.24 - Practice Test for Objects
    - Test yourself, avoid using outside resources aside from the reference sheet.
    - Skip 1.25

Friday 9/19

- 1.26 - Unit 1 FRQ
    - Test yourself, avoid using outside resources aside from the reference sheet.

Week 7 - Intro to Logic

Monday 9/22

- 2.1 - Algorithms with Selection and Repetition
    - Take detailed notes, focus on activities.

Tuesday 9/23

- 2.1 - Algorithms with Selection and Repetition
    - Take detailed notes, focus on activities.

Wednesday 9/24

- 2.2 - Boolean Expressions
    - Take detailed notes, focus on activities.

Thursday 9/25

- 2.3 - if Statements
    - Take detailed notes, focus on activities.

Friday 9/26

- 2.3 - if Statements
    - Take detailed notes, focus on activities.

Week 8 - More on Logic and Iteration

Monday 9/29

- 2.4 - Nested if statements
    - Take detailed notes, focus on activities.

Tuesday 9/30

- 2.4 - Nested if statements
    - Take detailed notes, focus on activities.

Wednesday 10/1

- 2.5 - Compound Boolean Expressions
    - Take detailed notes, focus on activities.

Thursday 10/2

- 2.5 - Compound Boolean Expressions
    - Take detailed notes, focus on activities.

Friday 10/3

- 2.6 - Comparing Boolean Expressions (De Morgan's Laws)
    - Take detailed notes, focus on activities.

Week 9 - Definite and Indefinite Iteration

Monday 10/6

- 2.6 - Comparing Boolean Expressions (De Morgan's Laws)
    - Take detailed notes, focus on activities.

Tuesday 10/7

- 2.7 - While Loops
    - Take detailed notes, focus on activities.

Wednesday 10/8

- 2.7 - While Loops
    - Take detailed notes, focus on activities.

Thursday 10/9

- 2.8 - For Loops
    - Take detailed notes, focus on activities.

Friday 10/10

- 2.8 - For Loops
    - Take detailed notes, focus on activities.

Week 10 - More on Iteration - Integration with Algorithms

Monday 10/13

- 2.9 - Implementing Selection & Iteration Algorithms
    - Take detailed notes, focus on activities.
    - How do you use loops in algorithms to complete tasks?

Tuesday 10/14

- 2.9 - Implementing Selection & Iteration Algorithms
    - Take detailed notes, focus on activities.
    - What does the accumulator pattern accomplish/do?

Wednesday 10/15

- 2.9 - Implementing Selection & Iteration Algorithms // 2.10 - Implementing String Algorithms
    - Take detailed notes, focus on activities.
    - Review how these concepts are integrated, begin 2.10 after completion

Thursday 10/16

- 2.10 - Implementing String Algorithms
    - Take detailed notes, focus on activities.

Friday 10/17

- 2.10 - Implementing String Algorithms
    - Take detailed notes, focus on activities.
    - Short review of 2.9 & 2.10 notes (optional)

Week 11 - Review for Unit 2

Monday 10/20 - Delayed Start

- 2.11 - Nested Iteration
    - Take detailed notes, focus on activities.

Tuesday 10/21

- 2.11 - Nested Iteration
    - Take detailed notes, focus on activities.
    - How do you use loops in algorithms to complete tasks?

Wednesday 10/22

- 2.12 - Informal Runtime Analysis of Loops

Thursday 10/23

- Review Day
    - Catch up on work.
    - Review Unit 2 through 2.13-2.23 if necessary.

Friday 9/24

- 2.24 - Unit 2 Test
    - Test yourself, avoid using outside resources aside from the reference sheet.

Week 12 - Intro to Classes and Object-Oriented Programming

Monday 10/27

- 3.1 - Abstraction and Program Design
    - Take detailed notes, focus on activities.

Tuesday 10/28

- 3.2 - Impact of Program Design
    - Take detailed notes, focus on activities.

Wednesday 10/29

- 3.3 - Anatomy of a Java Class
    - Important topic!
    - Take detailed notes, focus on activities.

Thursday 10/30

- 3.3 - Anatomy of a Java Class
    - Important topic!
    - Take detailed notes, focus on activities.

Friday 10/31

- No School

Week 13 - Intro to Classes and Object-Oriented Programming

Monday 11/3

- 3.3 - Anatomy of a Java Class
    - Important topic!
    - Take detailed notes, focus on activities.

Tuesday 11/4

- 3.4 - Writing Constructors
    - Take detailed notes, focus on activities.

Wednesday 11/5

- 3.5 - Methods: How to Write Them
    - Another important topic!
    - Take detailed notes, focus on activities.

Thursday 11/6

- 3.5 - Methods: How to Write Them
    - Take detailed notes, focus on activities.

Friday 11/7

- 3.6 - Methods: Passing and Returning References on Objects
    - Take detailed notes, focus on activities

Week 14 - Break

Twelve weeks have passed, that is enough for a term. Please use this week to catch up on other schoolwork as things ramp up for the final grading period. Best of luck!

Monday 11/10

- No notes.

Tuesday 11/11 - No School

- No notes.

Wednesday 11/12

- No notes.

Thursday 11/13

- No notes.

Friday 11/14

- No notes.

Week 15 - Class Variables

Monday 11/17

- 3.7 - Class (static) Variables and Methods
    - Take detailed notes, focus on activities.

Tuesday 11/18

- 3.8 - Scope and Access
    - Take detailed notes, focus on activities.

Wednesday 11/19

- 3.9 - this Keyword
    - Take detailed notes, focus on activities.

Thursday 11/20

- 3.10 - Unit 3 Summary
    - Take detailed notes, focus on activities.

Friday 11/21

- 3.10 - Unit 3 Summary
    - Take detailed notes, focus on activities.

Week 16 - Thanksgiving Break

Monday 11/24

- Thanksgiving break.

Tuesday 11/25

- Thanksgiving break.

Wednesday 11/26

- Thanksgiving break.

Thursday 11/27

- Thanksgiving break.

Friday 11/28

- Thanksgiving break.

Week 17 - Introduction to Data Collection

Monday 12/1

- 4.1 - Ethical and Social Issues Around Data Collection

    - Take detailed notes, focus on activities.

Tuesday 12/2

- 4.2 - Data Sets
    - Take detailed notes, focus on activities.

Wednesday 12/3

- 4.2 - Data Sets
    - Take detailed notes, focus on activities.

Thursday 12/4

- 4.3 - Array Creation and Access
    - Key topic! Take notes.

Friday 12/5

- 4.3 - Array Creation and Access
    - Key topic! Take notes and review.

Week 18 - More on Arrays

Monday 12/8

- 4.4 - Array Traversals
    - Take detailed notes, focus on activities.

Tuesday 12/9

- 4.4 - Array Traversals
    - Take detailed notes, focus on activities.

Wednesday 12/10

- 4.5 - Implementing Array Algorithms
    - Take detailed notes, focus on activities.

Thursday 12/11

- 4.5 - Implementing Array Algorithms
    - Take detailed notes, focus on activities.

Friday 12/12

- 4.6 - Using Text Files

  - Take detailed notes, focus on activities.

Week 19 - Arrays

Return to school + week breaks because we were lazy. Start in the second week of February.

Monday 2/9

- 4.7 - Wrapper Classes - Integer and Double

Tuesday 2/10

- 4.8 - ArrayList and its Methods

Wednesday 2/11

- 4.8 - ArrayList and its Methods

Thursday 2/12

- 4.9 - ArrayList Traversals

Friday 2/13

- 4.9 - ArrayList Traversals

Week 20 - Arrays

Monday 2/16

- No School

Tuesday 2/17

- 4.10 - Implementing ArrayList Algorithms

Wednesday 2/11

- 4.11 - 2D Array Creation and Access

Thursday 2/12

- 4.11 - 2D Array Creation and Access

Friday 2/13

- No School

Week 21 - More on Logic and Algorithms

Monday 2/16

- 4.12 - 2D Array Traversals: Nested Loops

Tuesday 2/17

- 4.12 - 2D Array Traversals: Nested Loops

Wednesday 2/18

- 4.13 - Implementing 2D Array Algorithms

Thursday 2/19

- 4.13 - Implementing 2D Array Algorithms

Friday 2/20

- 4.14 - Searching Algorithms

Week 22 - Recursion

Monday 2/23

- 4.15 - Sorting Algorithms

Tuesday 2/24

- 4.15 - Sorting Algorithms

Wednesday 2/25

- 4.16 - Recursion

Thursday 2/26

- 4.16 - Recursion

Friday 2/27

- No school.

Week 23 - More on Recursion and Review

Monday 3/2

- 4.17 - Recursive Searching and Sorting

Tuesday 3/3

- 4.17 - Recursive Searching and Sorting
    - This concludes AP CSA content!

Wednesday 3/4

- **First day of review.** This review schedule may depend on your available resources. While we recommend buying the Barron's Book for AP CSA (make sure it's at least after the 2025-2026 revisions), there are still available resources through both csawesome2 and AP CSA FRQs. Try to devote more time towards the latter units. Good luck!

Exam Day *Friday 5/15*

- Good luck! Don't stress out too much. Chances are if you've been following the material dutifully, reviewing through FRQs and CSAwesome, you've already done more than enough.